



Department of Electrical and Instrumentation Engineering
TIET

Course Code: UBM702

Learning Objectives

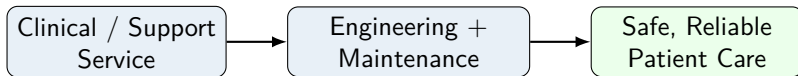
By the end of this lecture, students should be able to:

- ▶ explain the organization and responsibilities of a hospital engineering department;
- ▶ distinguish corrective, preventive and condition-based maintenance strategies;
- ▶ develop a risk-based maintenance programme for hospital equipment and utilities;
- ▶ interpret work orders, equipment histories and key maintenance indicators;
- ▶ calculate basic reliability measures such as MTBF, MTTR and availability;
- ▶ explain the biomedical engineer's role in maintenance, calibration, safety and replacement planning.

Connection with the Previous Lecture

Lecture 10 Established

CSSD safety depends on reliable sterilizers, washer-disinfectors, water treatment, steam, power, ventilation and traceability systems.



Key Transition

Today we study the department that keeps hospital infrastructure and technology available throughout their life cycle.

Opening Scenario: 2:00 AM in the ICU

A ventilator suddenly stops delivering the set tidal volume while the patient is critically ill.

Immediate Questions

1. Who responds first?
2. Is a backup ventilator available?
3. Can the fault be isolated quickly?
4. Was preventive maintenance completed?
5. Does the engineer have the service history and spare parts?
6. Should this event trigger wider inspection of similar ventilators?

Maintenance is not only repair; it is continuity of care.

What Is a Hospital Engineering Department?

A hospital engineering department plans, operates, maintains and improves the technical systems required for healthcare delivery.

- ▶ building and facility services;
- ▶ electrical distribution and emergency power;
- ▶ HVAC and environmental control;
- ▶ water, sanitation and drainage;
- ▶ medical gas and vacuum systems;
- ▶ fire and life-safety systems;
- ▶ medical equipment and clinical technology;
- ▶ communication, monitoring and building-management systems.

Why Maintenance Matters in a Hospital

Clinical Consequences

- ▶ delayed diagnosis or treatment;
- ▶ cancelled procedures;
- ▶ unsafe equipment performance;
- ▶ infection-control failure;
- ▶ loss of critical utilities.

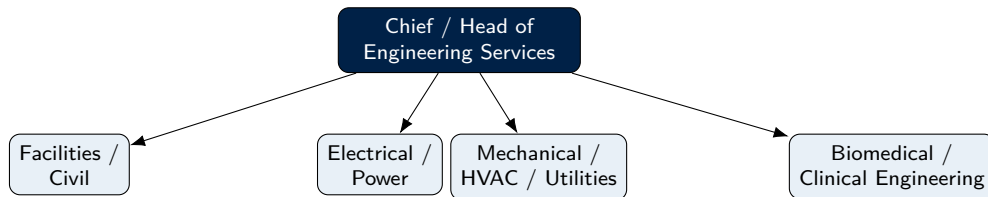
Organizational Consequences

- ▶ downtime and lost capacity;
- ▶ emergency repair cost;
- ▶ staff dissatisfaction;
- ▶ regulatory non-compliance;
- ▶ shortened asset life.

Major Functions of the Engineering Department

1. technical planning and design coordination;
2. installation and commissioning;
3. asset registration and documentation;
4. preventive and corrective maintenance;
5. calibration and safety testing;
6. utility operation and monitoring;
7. energy and resource management;
8. spare-parts and contract management;
9. incident investigation and risk reduction;
10. replacement and technology-upgrade planning.

Typical Engineering Department Structure



Important

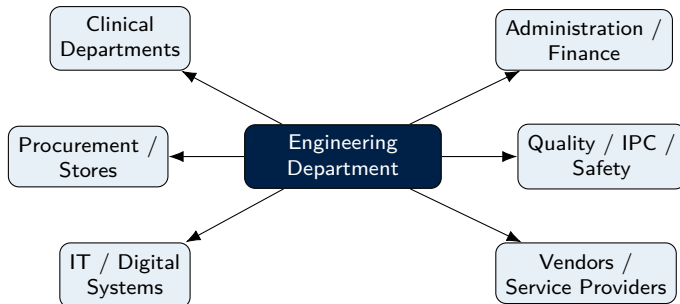
Exact organization varies with hospital size, outsourcing model and available technical staff.

Facility Engineering vs Clinical Engineering

Area	Facility Engineering	Biomedical / Clinical Engineering
Primary focus	Infrastructure and utilities	Medical technology
Examples	Power, HVAC, water, lifts, fire systems	Ventilators, monitors, defibrillators, imaging support
Maintenance	Plant and building systems	Medical equipment lifecycle
Safety	Electrical, mechanical, fire, environmental	Electrical safety, calibration, device performance
Users	Entire facility	Clinical departments

Both groups must coordinate for safe clinical care.

The Engineering Department Is Highly Interconnected



Asset Life Cycle



Engineering View

Maintenance begins before the first breakdown: good specification, installation and user training determine future reliability.

Asset Inventory: The Starting Point

A maintenance programme cannot be controlled without knowing what the hospital owns.

- ▶ unique asset ID;
- ▶ equipment/system name;
- ▶ manufacturer, model and serial number;
- ▶ location and user department;
- ▶ purchase and commissioning dates;
- ▶ warranty / AMC / CMC status;
- ▶ preventive-maintenance frequency;
- ▶ calibration requirement;
- ▶ risk/criticality category;
- ▶ current service status.

Asset Identification and Labelling

A Useful Asset Label May Contain

- ▶ hospital asset number;
- ▶ barcode or QR code;
- ▶ department / location;
- ▶ PM due date;
- ▶ calibration status where applicable.

Digital Link

Scanning the code can open:

- ▶ equipment history;
- ▶ user manual;
- ▶ service manual;
- ▶ open work orders;
- ▶ PM checklist;
- ▶ spare-parts history.

Not Every Asset Has the Same Risk

Maintenance priority should reflect consequences of failure.

Possible Criticality Factors

- ▶ direct effect on patient life or diagnosis;
- ▶ probability of failure;
- ▶ availability of backup equipment;
- ▶ complexity of detection before harm;
- ▶ intensity/frequency of clinical use;
- ▶ regulatory or calibration requirement;
- ▶ repair lead time and spare-parts availability.

Simple Risk-Based Prioritization

Asset	Consequence	Backup	Priority
Ventilator	Very high	Limited	Very high
Defibrillator	Very high	Limited	Very high
Infusion pump	High	Often available	High
Patient monitor	High	Variable	High
Wheelchair	Moderate	Usually available	Moderate
Office printer	Low	Easy alternative	Low

Engineering Principle

Risk-based maintenance directs limited technical resources toward assets whose failure matters most.

Maintenance Philosophy

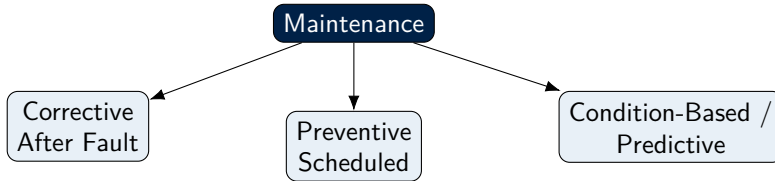
The objective is not to eliminate every failure at any cost.

$$\text{Maintenance} = \text{Appropriate Reliability} + \text{Safety} + \text{Availability} + \text{Cost Control}$$

A good programme balances:

- ▶ patient risk;
- ▶ equipment criticality;
- ▶ manufacturer recommendations;
- ▶ observed failure history;
- ▶ operating environment;
- ▶ technical resources and budget.

Major Maintenance Strategies



Corrective or Breakdown Maintenance

Corrective maintenance restores function after a fault is identified.

- ▶ complaint received or fault detected;
- ▶ equipment made safe and isolated if necessary;
- ▶ problem diagnosed;
- ▶ component repaired/replaced;
- ▶ function and safety verified;
- ▶ work documented and equipment returned to service.

Limitation

A purely reactive strategy is unsuitable for safety-critical assets because failure has already occurred before action starts.

Preventive Maintenance (PM)

PM is planned maintenance performed at predetermined intervals to reduce the probability of failure or degradation.

- ▶ visual inspection;
- ▶ cleaning and lubrication;
- ▶ filters, seals or batteries as required;
- ▶ functional checks;
- ▶ alarm verification;
- ▶ electrical-safety checks where applicable;
- ▶ calibration or performance verification where required;
- ▶ documentation of findings and next due date.

Condition-Based and Predictive Maintenance

Maintenance is triggered by observed equipment condition rather than calendar time alone.

Examples

- ▶ vibration trend of pumps or compressors;
- ▶ temperature trend of bearings;
- ▶ battery capacity trend;
- ▶ filter differential pressure;
- ▶ error-log frequency;
- ▶ sterilizer cycle deviations;
- ▶ power-quality events or motor-current signatures.

Detect deterioration before functional failure.

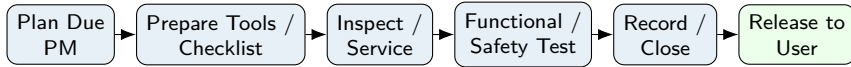
Maintenance Is Not the Same as Calibration

	Maintenance	Calibration / Verification
Purpose	Keep or restore function	Confirm measurement/performance accuracy
Typical action	Clean, inspect, repair, replace	Compare against known reference
Outcome	Equipment operates correctly	Error is quantified and within acceptance limit
Example	Replace worn NIBP cuff hose	Verify pressure measurement using simulator

Key Point

Some maintenance visits include calibration checks, but the two activities have different objectives.

Preventive Maintenance Workflow



How Is a PM Interval Selected?

Possible inputs include:

- ▶ manufacturer recommendation;
- ▶ equipment criticality;
- ▶ legal or accreditation requirement;
- ▶ utilization intensity;
- ▶ age and failure history;
- ▶ operating environment;
- ▶ findings from earlier PM visits;
- ▶ institutional experience and risk assessment.

Avoid

Do not copy the same PM interval blindly to every device category.

Example PM Plan: ICU Ventilator

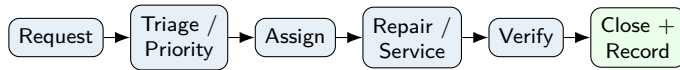
Step	Typical Verification
Visual inspection	Housing, cables, tubing, connectors, wheels, labels
Power system	AC operation, battery condition, charging
Ventilation	Volume/pressure delivery using appropriate analyser
Alarms	High/low pressure, apnea, disconnect, oxygen-related alarms as applicable
Sensors	Flow/pressure/O ₂ sensor performance as applicable
Safety	Leakage/grounding checks where required
Documentation	Results, parts changed, due date, technician

Work Order: The Unit of Maintenance Control

Every significant maintenance activity should be traceable through a work order.

- ▶ requester and department;
- ▶ asset ID and location;
- ▶ problem description;
- ▶ priority / response category;
- ▶ date and time opened;
- ▶ technician assigned;
- ▶ diagnosis and action taken;
- ▶ parts and labour used;
- ▶ downtime;
- ▶ verification and closure;
- ▶ user acceptance where applicable.

Work Order Life Cycle



Computerized Maintenance Management System (CMMS)

A CMMS centralizes maintenance information and workflow.

Core Functions

- ▶ asset register;
- ▶ work orders;
- ▶ PM scheduling;
- ▶ parts inventory;
- ▶ contracts and warranties;
- ▶ equipment history.

Management Outputs

- ▶ overdue PM list;
- ▶ downtime report;
- ▶ failure trends;
- ▶ cost by device type;
- ▶ technician workload;
- ▶ replacement candidates.

Equipment History Is Valuable Data

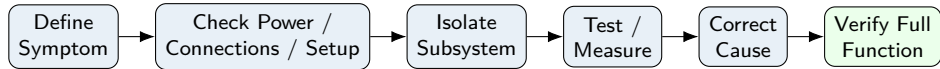
For each asset, the maintenance history can reveal:

- ▶ repeated component failures;
- ▶ increasing repair frequency;
- ▶ high downtime;
- ▶ poor vendor support;
- ▶ costly spare parts;
- ▶ user-related damage patterns;
- ▶ model-wide reliability problems;
- ▶ when replacement may be more economical than repair.

Corrective Maintenance Workflow

1. make the situation clinically safe;
2. identify and label the equipment;
3. reproduce/confirm the fault where safe;
4. check simple causes first;
5. consult logs, manuals and history;
6. isolate subsystem and diagnose root cause;
7. repair or replace component;
8. perform functional and safety verification;
9. document the work and release equipment;
10. assess whether wider preventive action is needed.

A Structured Troubleshooting Logic



Common Failure Categories in Medical Equipment

- ▶ power-supply or battery failure;
- ▶ sensor degradation;
- ▶ cables, connectors and accessories;
- ▶ pumps, valves and mechanical wear;
- ▶ contamination or fluid ingress;
- ▶ software/configuration problems;
- ▶ user setup or accessory mismatch;
- ▶ environmental conditions;
- ▶ consumable-related faults;
- ▶ component ageing.

Root-Cause Analysis: Ask Beyond the Failed Part

Example

An infusion pump battery repeatedly fails after only a few months.

1. Why did the battery lose capacity? – It remained deeply discharged.
2. Why? – Pumps were often stored unplugged.
3. Why? – Charging points were insufficient in the storage area.
4. Why? – Storage was planned without electrical-load assessment.
5. Why? – Engineering was not involved in workflow planning.

The root cause may be a system-design problem, not only a component failure.

Failure Mode and Effects Analysis (FMEA)

FMEA is a proactive method used to identify how a process or system might fail before harm occurs.

- ▶ define process or asset;
- ▶ identify failure modes;
- ▶ identify causes and effects;
- ▶ estimate severity and likelihood;
- ▶ assess detectability where used;
- ▶ prioritize higher-risk failure modes;
- ▶ design controls;
- ▶ reassess after improvement.

Illustrative Risk Priority Number

One common teaching approach is:

$$RPN = S \times O \times D$$

where:

- ▶ S = severity score;
- ▶ O = occurrence score;
- ▶ D = detectability score.

Example

If $S = 9$, $O = 4$, $D = 5$, then $RPN = 180$.

Caution

RPN is a prioritization aid, not a substitute for clinical judgement; a very severe hazard may require action even with a lower total score.

Reliability Concepts for Maintenance Management

- ▶ **Reliability:** probability of performing required function for a specified period under stated conditions;
- ▶ **Maintainability:** ease and speed with which an asset can be restored;
- ▶ **Availability:** proportion of time the asset is ready for use;
- ▶ **Downtime:** period during which the asset is unavailable for required service.

Mean Time Between Failures (MTBF)

For a repairable asset, a simple operational estimate is:

$$\text{MTBF} = \frac{\text{Total operating time}}{\text{Number of failures}}$$

Example

A ventilator fleet accumulates 12,000 operating hours with 6 relevant failures.

$$\text{MTBF} = \frac{12,000}{6} = 2,000 \text{ hours}$$

Higher MTBF generally indicates longer operation between failures, provided the data are defined consistently.

Mean Time to Repair (MTTR)

A simple estimate is:

$$\text{MTTR} = \frac{\text{Total corrective repair time}}{\text{Number of repairs}}$$

Example

Five repairs required a total of 15 technician-hours.

$$\text{MTTR} = \frac{15}{5} = 3 \text{ hours/repair}$$

Reducing MTTR may require better diagnostics, spares, documentation, access and training.

Availability

For a simple repairable system model:

$$A \approx \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

Example

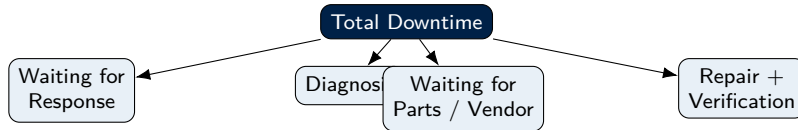
If MTBF = 2,000 h and MTTR = 3 h:

$$A = \frac{2000}{2000 + 3} \approx 0.9985 \approx 99.85\%.$$

Remember

Operational availability may also be reduced by waiting for approvals, vendors, spare parts or transport, not only hands-on repair time.

Where Does Downtime Come From?



Management Opportunity

Often the largest reduction in downtime comes from workflow and logistics, not faster soldering.

Useful Maintenance Key Performance Indicators

- ▶ preventive-maintenance compliance;
- ▶ overdue PM count;
- ▶ equipment downtime;
- ▶ MTBF and MTTR trends;
- ▶ repeat-repair rate;
- ▶ first-time-fix rate;
- ▶ response time for critical calls;
- ▶ spare-parts lead time;
- ▶ maintenance cost per asset / device category;
- ▶ percentage of assets under active service coverage.

Preventive-Maintenance Compliance

A simple indicator is:

$$\text{PM Compliance} = \frac{\text{PM jobs completed on time}}{\text{PM jobs due}} \times 100\%$$

Example

If 186 of 200 scheduled PM jobs were completed on time:

$$\text{PM Compliance} = \frac{186}{200} \times 100 = 93\%.$$

The metric should be interpreted together with asset criticality and reasons for delay.

First-Time-Fix Rate

$$\text{FTF} = \frac{\text{Repairs closed without repeat visit}}{\text{Total repairs}} \times 100\%$$

A low first-time-fix rate may suggest:

- ▶ incomplete diagnosis;
- ▶ inadequate spares;
- ▶ limited training;
- ▶ poor technical documentation;
- ▶ intermittent faults;
- ▶ premature closure of work orders.

Maintenance Cost: Look Beyond Repair Bills

Total ownership cost may include:

- ▶ purchase and installation;
- ▶ annual service contracts;
- ▶ preventive-maintenance labour;
- ▶ spare parts and consumables;
- ▶ calibration and quality-control tools;
- ▶ utilities and infrastructure;
- ▶ software/licensing;
- ▶ downtime and lost clinical capacity;
- ▶ training;
- ▶ disposal and replacement.

Repair or Replace?

Replacement becomes more attractive when several factors combine:

- ▶ frequent failures;
- ▶ increasing downtime;
- ▶ high repair cost;
- ▶ unavailable spare parts;
- ▶ manufacturer support ending;
- ▶ poor clinical performance;
- ▶ safety or compliance concerns;
- ▶ inability to integrate with current systems;
- ▶ better lifecycle economics of newer technology.

A Simple Replacement Decision Matrix

Criterion	Good	Moderate	Poor
Reliability	Low failures	Rising failures	Frequent failures
Support	Strong	Limited	Obsolete
Safety	Fully acceptable	Minor concerns	Significant concern
Repair cost	Low	Moderate	High
Clinical value	High	Adequate	Low / outdated

Replacement decisions should use evidence from the maintenance history.

Spare-Parts Management

Poor spare-part planning converts a short repair into long downtime.

- ▶ identify critical fast-moving parts;
- ▶ maintain minimum/maximum stock levels;
- ▶ control shelf life where relevant;
- ▶ record part number and compatible models;
- ▶ track issue against work order;
- ▶ avoid uncontrolled cannibalization;
- ▶ monitor lead time and vendor availability;
- ▶ review obsolete or rarely used inventory.

AMC and CMC

Annual Maintenance Contract (AMC)

Typically provides defined service visits and labour support; parts coverage depends on the contract.

Comprehensive Maintenance Contract (CMC)

Typically includes broader repair support and specified parts, subject to exclusions and contract terms.

Contract evaluation should examine: response time, uptime commitment, PM frequency, exclusions, parts coverage, software support, documentation and escalation process.

Vendor Performance Should Be Measured

Possible vendor indicators include:

- ▶ response time;
- ▶ repair turnaround time;
- ▶ repeat failure after service;
- ▶ PM completion;
- ▶ spare-part lead time;
- ▶ documentation quality;
- ▶ training support;
- ▶ uptime achievement;
- ▶ user/engineering satisfaction;
- ▶ contract compliance.

Warranty Management

Engineering should know:

- ▶ warranty start and end dates;
- ▶ what is included/excluded;
- ▶ preventive-maintenance obligations;
- ▶ whether unauthorized repair voids warranty;
- ▶ who must be contacted for service;
- ▶ response-time commitments;
- ▶ replacement/loaner provisions;
- ▶ software and accessory coverage.

Common Loss

Hospitals sometimes pay for repairs that should have been covered because warranty information is not visible to the technician.

Acceptance Testing and Commissioning

Before new equipment enters clinical service, engineering should confirm:

- ▶ delivered model and configuration match the purchase order;
- ▶ accessories and documentation are complete;
- ▶ utilities and site conditions are suitable;
- ▶ equipment starts and performs as expected;
- ▶ safety checks are completed as applicable;
- ▶ network/interface requirements are verified;
- ▶ user and technical training are arranged;
- ▶ asset ID and maintenance schedule are created;
- ▶ warranty is recorded.

Electrical Safety in Maintenance

Biomedical maintenance may include appropriate checks of:

- ▶ mains cord and plug condition;
- ▶ protective earth continuity where applicable;
- ▶ insulation or leakage-current performance where required;
- ▶ accessible conductive parts;
- ▶ correct fuse rating;
- ▶ battery condition;
- ▶ safe function after repair.

Important

Tests and acceptance limits depend on equipment class, applied parts, applicable standards and institutional procedures.

Calibration and Measurement Traceability

Examples of devices requiring performance verification may include:

- ▶ infusion/syringe pumps;
- ▶ NIBP monitors;
- ▶ ventilators;
- ▶ defibrillators;
- ▶ weighing scales;
- ▶ temperature-monitoring systems;
- ▶ laboratory analysers;
- ▶ radiation/imaging quality-control instruments.

Traceability

Reference test equipment itself must be controlled and calibrated according to the hospital's quality system.

Maintenance in Critical Clinical Areas

Maintenance planning is more demanding in:

- ▶ ICU and NICU;
- ▶ operation theatres;
- ▶ emergency department;
- ▶ dialysis;
- ▶ radiology and imaging;
- ▶ laboratory;
- ▶ CSSD;
- ▶ oxygen/manifold and other critical utility areas.

Work may require backup equipment, infection-control precautions, permit systems and planned clinical downtime.

Utility Maintenance: Electrical System

Critical electrical maintenance may include:

- ▶ transformers and switchgear;
- ▶ distribution boards;
- ▶ generators and automatic transfer systems;
- ▶ UPS systems and batteries;
- ▶ earthing and bonding;
- ▶ emergency lighting;
- ▶ power-quality monitoring;
- ▶ thermal inspection of connections where appropriate.

Utility Maintenance: HVAC

Important maintenance tasks may include:

- ▶ filters and differential-pressure monitoring;
- ▶ fan and motor condition;
- ▶ coils and drain pans;
- ▶ temperature and humidity sensors;
- ▶ pressure relationships in critical rooms;
- ▶ air-volume / airflow verification;
- ▶ control-system alarms;
- ▶ cleanliness and access for safe servicing.

Utility Maintenance: Medical Gas Systems

Maintenance attention includes:

- ▶ source equipment and manifolds;
- ▶ compressors and vacuum pumps;
- ▶ alarms;
- ▶ zone valves;
- ▶ pressure regulators;
- ▶ terminal outlets;
- ▶ pipeline integrity;
- ▶ reserve supply and emergency procedures.

Critical Principle

Any intervention on a live medical-gas system requires strict control because an error can affect multiple patients simultaneously.

Utility Maintenance: Water and Sanitation

Engineering may be responsible for:

- ▶ pumps and storage tanks;
- ▶ hot-water systems;
- ▶ water treatment;
- ▶ dialysis-related utility interfaces where applicable;
- ▶ plumbing and drainage;
- ▶ sewage systems;
- ▶ leak detection;
- ▶ temperature/quality monitoring as required;
- ▶ coordination with infection-control teams.

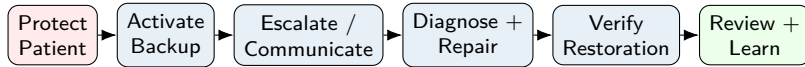
Planned Shutdowns Require Coordination

A planned shutdown should answer:

1. What system will be unavailable?
2. Which departments and patients are affected?
3. What backup is available?
4. When is the lowest-risk time?
5. Who authorizes the shutdown?
6. Who communicates start and restoration?
7. What test confirms safe return to service?
8. What is the contingency if work exceeds planned time?

Emergency Maintenance Response

For a critical failure:



Infection Control During Maintenance

Engineering work can create infection risk through dust, water, air disturbance and contaminated equipment.

- ▶ coordinate with infection-control team for high-risk work;
- ▶ isolate work area when required;
- ▶ control dust and debris;
- ▶ protect clean/sterile areas;
- ▶ decontaminate equipment before and after servicing as appropriate;
- ▶ use PPE based on hazard assessment;
- ▶ restore environmental controls before reopening area.

Software and Cybersecurity Are Maintenance Issues Too

Modern medical devices may depend on software, networks and connected services.

- ▶ maintain software/firmware inventory;
- ▶ coordinate approved updates;
- ▶ protect service credentials;
- ▶ preserve configuration backups;
- ▶ document network changes;
- ▶ coordinate with hospital IT/cybersecurity teams;
- ▶ avoid unsupported software changes during repair;
- ▶ plan for vendor support and obsolescence.

When Maintenance Becomes a Safety Incident

Examples include:

- ▶ equipment returned without complete functional verification;
- ▶ wrong replacement part installed;
- ▶ calibration error left undetected;
- ▶ utility shutdown performed without clinical notification;
- ▶ alarm disabled during service and not restored;
- ▶ service setting changed inadvertently;
- ▶ contaminated equipment moved into workshop without precautions.

Response

Make safe, report, investigate, correct the system weakness and communicate learning.

Case Study 1: Repeated Ventilator Failure

Three ventilators of the same model show intermittent “low pressure” alarms over two months.

Discuss

1. Should each event be treated as an isolated repair?
2. What data should be extracted from the CMMS?
3. Could accessories, maintenance practice or environment be contributing?
4. Should all units of the model be inspected?
5. When should the manufacturer be contacted?

Maintenance data can reveal a fleet-level problem before a serious event occurs.

Case Study 2: Autoclave Downtime

A CSSD steam sterilizer fails repeatedly because of unstable steam quality. The sterilizer itself passes component checks.

Engineering Question

Is the sterilizer the true root cause?

Possible investigation:

- ▶ steam supply pressure and quality;
- ▶ boiler operation;
- ▶ traps and condensate system;
- ▶ distribution piping;
- ▶ water treatment;
- ▶ load pattern and operating procedure;
- ▶ cycle trend data.

Case Study 3: “No Fault Found” Infusion Pumps

Several pumps are repeatedly sent to engineering with the complaint “occlusion alarm,” but bench testing shows normal function.

Look Beyond the Device

Investigate:

- ▶ administration sets and accessories;
- ▶ tubing placement;
- ▶ clamp/line conditions;
- ▶ clinical setup and training;
- ▶ alarm-limit configuration;
- ▶ workflow and documentation;
- ▶ whether the fault is intermittent.

Classroom Activity: Calculate the Maintenance Indicators

For a group of patient monitors in one year:

- ▶ total operating time = 30,000 h;
- ▶ failures = 12;
- ▶ total active repair time = 36 h;
- ▶ PM jobs due = 80;
- ▶ PM jobs completed on time = 72.

Calculate

1. MTBF;
2. MTTR;
3. approximate availability using MTBF and MTTR;
4. PM compliance.

Classroom Activity: Build a PM Programme

Your hospital has:

- ▶ 10 ventilators;
- ▶ 25 infusion pumps;
- ▶ 8 defibrillators;
- ▶ 40 patient monitors;
- ▶ 3 portable ultrasound systems.

In Groups

Choose a risk category, proposed maintenance frequency, two key checks and backup strategy for each device group. Explain your reasoning.

Mini-Project Ideas

Possible B.Tech Biomedical Engineering projects:

- ▶ CMMS dashboard for biomedical equipment;
- ▶ risk-based preventive-maintenance prioritization tool;
- ▶ predictive battery-health model for medical devices;
- ▶ QR-code equipment-history system;
- ▶ spare-parts optimization using ABC/VED concepts;
- ▶ reliability analysis of a selected device fleet;
- ▶ maintenance KPI dashboard for ICU equipment;
- ▶ replacement-priority scoring model;
- ▶ anomaly detection from equipment error logs.

Key Engineering Questions for Any Maintenance Programme

Before accepting a maintenance system, ask:

1. Do we know every critical asset and its current status?
2. Are maintenance priorities linked to patient risk?
3. Is every intervention traceable?
4. Are PM jobs completed on time?
5. Are test instruments controlled and calibrated?
6. Are recurring failures analysed, not merely repaired?
7. Are spare parts, vendors and contracts managed using data?
8. Is replacement planning based on lifecycle evidence?

Summary

- ▶ Hospital engineering integrates facilities, utilities and medical technology.
- ▶ Maintenance is a patient-safety and service-continuity function, not simply a repair activity.
- ▶ Corrective, preventive and condition-based strategies should be selected according to risk and evidence.
- ▶ Asset inventory, work orders, CMMS and equipment histories provide the information needed for control.
- ▶ MTBF, MTTR, availability and PM compliance help quantify performance.
- ▶ Biomedical engineers support commissioning, PM, calibration, safety testing, fault analysis and replacement planning.

A reliable hospital is maintained by design, data and disciplined engineering practice.

Assignment for the Next Lecture

Select one hospital asset or equipment category and prepare a one-page maintenance plan.

Include:

1. asset description and clinical role;
2. risk/criticality classification;
3. proposed preventive-maintenance interval;
4. PM checklist with at least five checks;
5. calibration/safety tests if applicable;
6. two likely failure modes;
7. spare-part or backup requirement;
8. two KPIs to monitor maintenance performance.

Submission Format

One page with a simple maintenance flowchart or table.

Suggested Reading

- ▶ World Health Organization, *Medical Equipment Maintenance Programme Overview*, WHO Medical Device Technical Series.
- ▶ World Health Organization, *Introduction to Medical Equipment Inventory Management*, WHO Medical Device Technical Series.
- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*.
- ▶ Manufacturer service manuals, preventive-maintenance procedures and institutional engineering SOPs for the equipment under study.
- ▶ Applicable electrical-safety, calibration, facility and medical-equipment standards; always consult the current edition adopted by the institution.

Next Lecture: Hospital Electrical Systems and Essential Power

Thank You

Questions and Discussion